David Rețe

Robotics Software Engineer: Perception, Mapping & Sensor Fusion

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Robotics software engineer with 3 years building and field-deploying multi-sensor 3D mapping pipelines (C++/Python, ROS 2, Docker) for the Horizon Europe **SeaClear 2.0**. I own the stack from mapping and sensor-fusion code to live operation on autonomous vessels, deployed across four sea-trial campaigns in Croatia, France, Germany, and Spain.

EXPERIENCE

Robotics Software Engineer — Horizon Europe SeaClear 2.0 2023 – present
TUCN, ROCON Group (Robotics and Nonlinear Control) Cluj-Napoca, RO

- Built the core **C++/ROS 2 detection-to-map pipeline**: projects camera and sonar object detections into a 3D OctoMap via occlusion-aware ray-casting, clusters litter objects, and serves their GPS coordinates to TU München's grapple system for cross-team autonomous retrieval.
- Architected the mapping subsystem as a hybrid **C++/Python** ROS 2 graph — latency-critical ray-casting and OctoMap updates in C++, with detection, sensor-bridging, georeferencing, and operator-facing service nodes in Python (rclpy) for fast field iteration.
- Designed and deployed a multi-container ROS 2 Jazzy mapping stack orchestrated via Docker Compose, running live on a USV during sea trials in Croatia (Dubrovnik), France (Marseille), Germany (Hamburg), and Spain (Tarragona).
- Wrote **Python** evaluation and post-mission tooling (NumPy, rosbag2 replay, pymap3d ENU↔WGS84) to georeference detections and validate end-to-end mapping accuracy against GPS-surveyed ground truth.
- Integrated and debugged ROV sensors: Oculus M-series multibeam sonar, Waterlinked A50 DVL, IMU, SBL acoustic positioning, and IP camera, consuming EKF-fused odometry for mapping.
- Built the **perception and operator tooling**: Python ROS 2 inference nodes running ONNX models on live camera and sonar streams with an auto/manual detection-mode service, plus **Foxglove Studio** bridges, live voxel-map visualization, and image-annotation panels.
- Developed a **probabilistic sonar mapping and active search** framework: log-odds occupancy mapping of detections and an information-driven next-best-view planner with sonar footprint, geofence, and bathymetry-based safety constraints, validated in closed-loop simulation.
- Diagnosed and fixed field-critical failures during live sea trials, including a **TF frame race** that mirrored sonar detections, a **Fast DDS shared-memory transport** issue between Docker containers, and a bounding-box scaling bug in the detection post-processing.

EDUCATION

MSc in Systems Engineering 2022 – 2024
TUCN, Faculty of Automation and Computer Science (taught in English)

BSc in Systems Engineering 2018 – 2022
TUCN, Faculty of Automation and Computer Science (taught in English). Thesis: LQR control for stabilization and trajectory tracking of a BlueROV2, implemented and validated in MATLAB/Simulink.

PUBLICATIONS

- **David Rețe**, P. Boroș, L. Bușoniu. "Active Mapping of Underwater Litter Using Camera-Sonar Fusion." *2026 IEEE International Conference on Automation, Quality and Testing, Robotics (AQTR)*, Băile Felix, Romania, pp. 1–6. doi: 10.1109/AQTR70159.2026.11577897

TECHNICAL SKILLS

- **Languages:** modern C++ (17/20), Python (rclpy, NumPy, OpenCV, ONNX Runtime), MATLAB/Simulink, Bash
- **Robotics:** ROS 2 Jazzy, TF2, Fast DDS (Discovery Server), OctoMap, rosbag tooling, sensor fusion integration
- **Perception & 3D:** OpenCV, camera calibration and projection models, multibeam sonar processing
- **Control:** LQR, PID, EKF, model-based control and tuning
- **Systems & DevOps:** Linux, Docker, GitLab CI/CD, GStreamer/RTSP, remote field deployment and debugging
- **Visualization:** Foxglove Studio (custom panels and bridges), RViz
- **Hardware deployed:** Oculus M-series multibeam sonar, Waterlinked A50 DVL, SBL acoustic positioning, BlueROV2, MiniTortuga ROV

LANGUAGES

Romanian (native) · English (fluent)